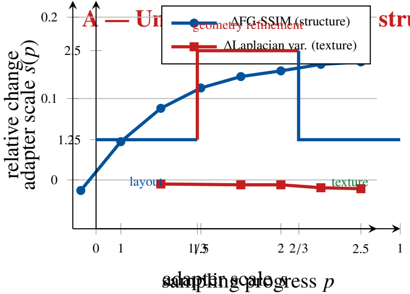
Timestep-Conditioned Adapter Scaling (TCAS) for Multi-view Diffusion Texture Generation

A — Ur… structure up, texture down



high scale ⇒ flat surfaces, detail loss

B — TCAS: stage-wise low–high–low schedule

C3 = (1.25, 2.50, 1.25): training-free, derived from measured stage utilities (CAI)

C — Validation: 300 objects, no re-search

+0.96 dB PSNR vs. aggressive uniform scaling s=2.50

FG-SSIM comparable to s=2.50 (n.s.)

Texture retention vs. s=1.25:
Lap. var. **83.5%**, RGB std **97.0%**, grad. **88.3%**
(s=2.50: 67.8%, 64.3%, 80.7%)

58.1% blinded overall first choices
(cluster bootstrap CI excludes zero)

TCAS: plug-and-play, no retraining, negligible overhead